

Digital Alarm Clock using Verilog HDL

1. Introduction

A Digital Alarm Clock is a digital system that keeps track of time and generates an alarm signal when the current time matches a predefined alarm time. This project was implemented using Verilog HDL and verified through simulation.

The project demonstrates the use of sequential logic, counters, clock management, and hardware verification techniques commonly used in digital system design.

2. Objective

The objective of this project is to design and simulate a digital alarm clock capable of:

- Tracking hours, minutes, and seconds
 - Displaying current time
 - Storing a predefined alarm time
 - Activating an alarm signal when the preset time is reached
 - Resetting the clock when required
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3. Tools Used

- Verilog HDL
 - EDA Playground
 - EPWave Waveform Viewer
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4. System Description

The Digital Alarm Clock operates using a clock signal and a set of counters that represent:

- Seconds
- Minutes
- Hours

The seconds counter increments with every clock cycle. After reaching its maximum value, it resets and increments the minutes counter. Similarly, the minutes counter increments the hours counter after completing a full cycle.

The current time is continuously compared with the preset alarm time. When both values match, the alarm output is activated.

5. Design Methodology

Step 1: Time Counting

The system maintains:

- Seconds Counter
- Minutes Counter
- Hours Counter

These counters operate synchronously with the clock signal.

Step 2: Alarm Storage

A predefined alarm time is stored within the system.

Step 3: Time Comparison

The current clock time is continuously compared with the stored alarm time.

Step 4: Alarm Activation

When the current time equals the alarm time, the alarm output becomes active.

Step 5: Reset Operation

The reset signal initializes all counters and alarm outputs to their default values.

6. Functional Blocks

Clock Divider

Generates timing signals suitable for clock operation.

Seconds Counter

Counts seconds and generates a carry signal after reaching its maximum count.

Minutes Counter

Counts minutes and increments after a complete second cycle.

Hours Counter

Tracks hours and resets after reaching the maximum hour value.

Alarm Comparator

Compares current time with alarm time.

Alarm Output Logic

Activates the alarm when a match occurs.

7. Simulation and Verification

The design was tested using simulation tools to verify:

- Correct time progression
- Proper counter rollover
- Alarm activation at the specified time
- Reset functionality
- Stable operation under continuous clock cycles

Waveform analysis was performed using EPWave.

8. Results

The simulation successfully demonstrated:

- Accurate timekeeping
- Correct hour, minute, and second counting
- Proper alarm triggering
- Successful reset operation

The generated waveforms confirmed that the design behaved according to the specified requirements.

9. Applications

- Digital clocks
 - Alarm systems
 - Embedded timing devices
 - FPGA-based digital systems
 - Educational digital electronics projects
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10. Learning Outcomes

This project provided practical experience in:

- Verilog HDL programming
 - Sequential circuit design
 - Counter implementation
 - Hardware simulation
 - Testbench development
 - Digital system verification
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11. Future Improvements

- Multiple alarm support
 - Snooze functionality
 - 12-hour and 24-hour modes
 - Seven-segment display integration
 - FPGA hardware implementation
 - Real-Time Clock (RTC) integration
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12. Conclusion

The Digital Alarm Clock was successfully designed and simulated using Verilog HDL. The project demonstrates fundamental digital design concepts including counters, sequential logic, and alarm generation mechanisms. It serves as a strong foundation for more advanced FPGA and digital system development projects.